

Objectives

- Explain React components
- Identify the differences between components and JavaScript functions
- Identify the types of components
- Explain class component
- Explain function component
- Define component constructor
- Define render() function

In this hands-on lab, you will learn how to:

- Create a function component
- Apply style to components
- Render a component

Prerequisites

The following is required to complete this hands-on lab:

- Node.js
- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **30 minutes.**

Create a react app for Student Management Portal named scorecalculatorapp and create a function component named "CalculateScore" which will accept Name, School, Total and goal in order to calculate the average score of a student and display the same.

1. Create a React project named "scorecalculatorapp" type the following command in terminal of Visual studio:

```
C:>npx create-react-app scorecalculatorapp
```

2. Create a new folder under Src folder with the name "Components". Add a new file named "CalculateScore.js"
3. Type the following code in CalculateScore.js

```

import './Stylesheets/mystyle.css'

const percentToDecimal= (decimal) => {
  return (decimal.toFixed(2) + '%')
}

const calcScore = (total, goal) => {
  return percentToDecimal(total/goal)
}

export const CalculateScore = ({Name,School, total, goal}) => (
  <div className="formatstyle">
    <h1><font color="Brown">Student Details:</font></h1>
    <div className="Name">
      <b> <span> Name: </span> </b>
      <span>{Name}</span>
    </div>
    <div className="School">
      <b> <span> School: </span> </b>
      <span>{School}</span>
    </div>
    <div className="Total">
      <b><span>Total:</span> </b>
      <span>{total}</span>
      <span>Marks</span>
    </div>
    <div className="Score">
      <b>Score:</b>
      <span>
        {calcScore(
          total,
          goal
        )}
      </span>
    </div>
  </div>
)

```

4. Create a Folder named Stylesheets and add a file named "mystyle.css" in order to add some styles to the components:

```

.Name
{
  font-weight:300;
  color:blue;
}
.School
{
  color:crimson;
}
.Total
{
  color:darkmagenta;
}
.formatstyle
{
  text-align:center;
  font-size:large;
}
.Score
{
  color:forestgreen;
}

```

5. Edit the App.js to invoke the CalculateScore functional component as follows:

```

import {CalculateScore} from '../src/components/CalculateScore';

function App()
{

return(
<div>
  <CalculateScore Name={"Steeve"}
    School={"DNV Public School"}
    total={284}
    goal={3}
  />
</div>
)
}

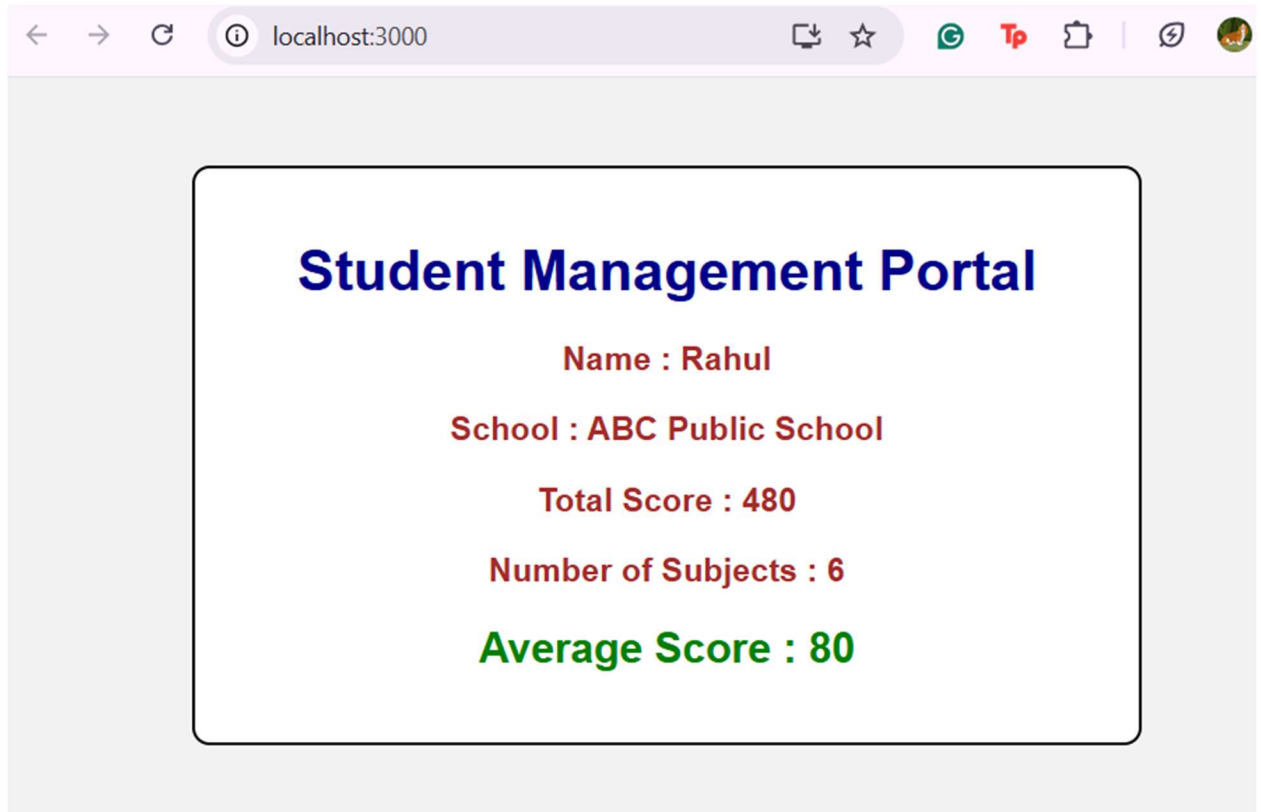
export default App;

```

6. In command Prompt, navigate into scorecalculatorapp and execute the code by typing the following command:

```
C:\scorecalculatorapp>npm start
```

7. Open browser and type "localhost:3000" in the address bar:



A **React component** is a reusable and independent piece of UI that returns JSX (JavaScript XML). Components help divide the user interface into smaller, manageable, and reusable parts.

Features

- Reusable
- Independent
- Accepts data through **props**
- Can manage its own **state**
- Returns JSX to display UI

There are two types of React components:

1. Function Component
2. Class Component

A Class Component is created using an ES6 class that extends `React.Component`.

Features

- Supports state
- Uses lifecycle methods
- Must contain a `render()` method
- More commonly used before Hooks were introduced

A Function Component is a simple JavaScript function that returns JSX.

Features

- Easy to write
- Lightweight
- Uses Hooks for state and lifecycle
- Most commonly used in modern React